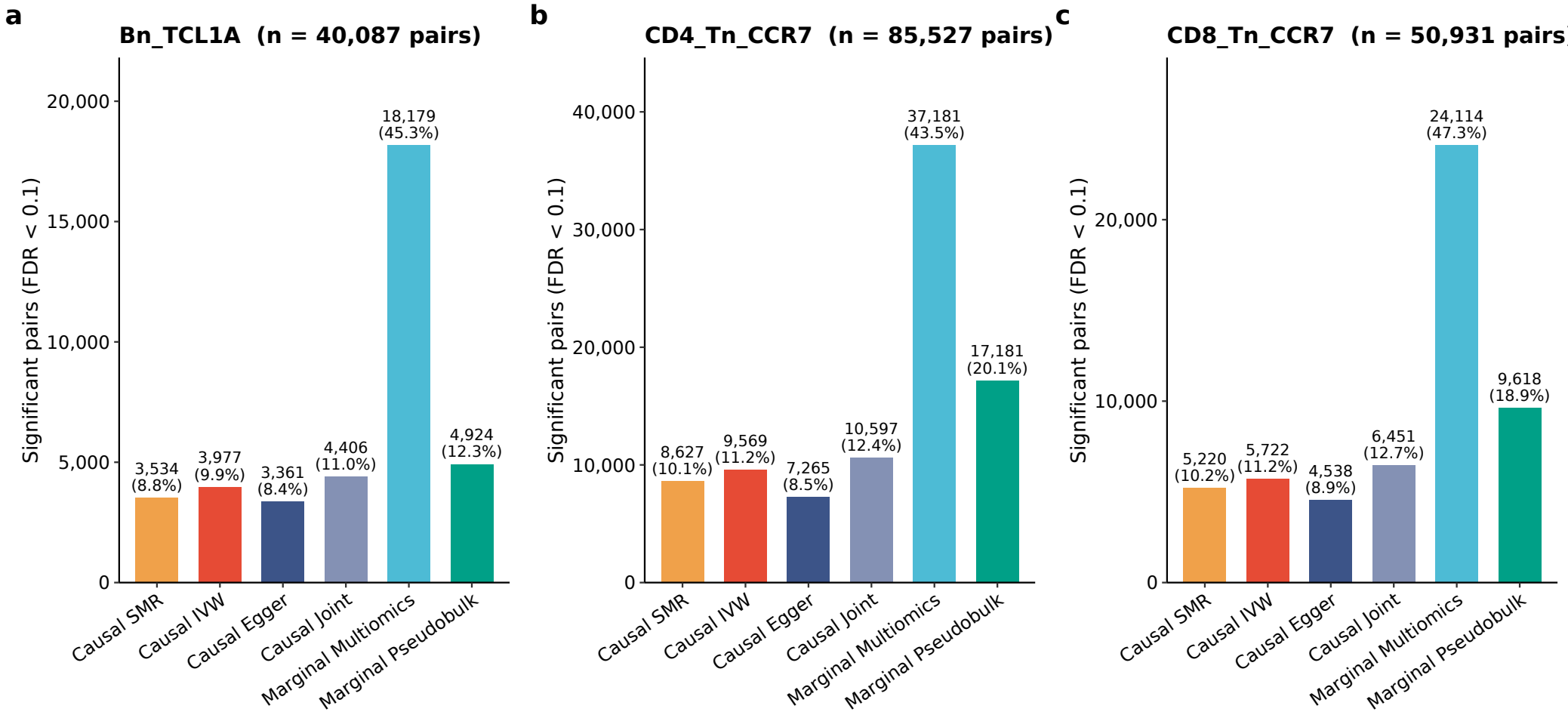
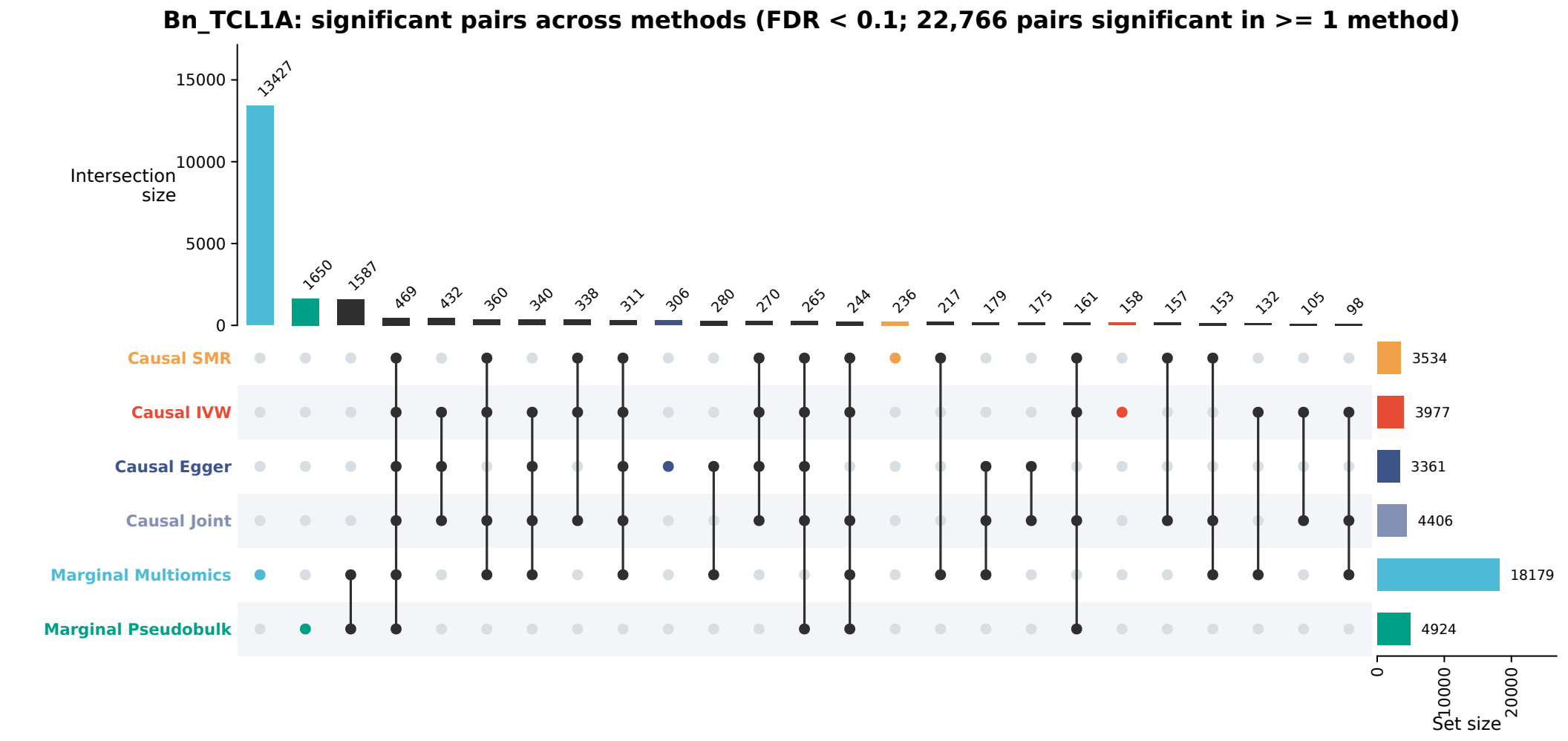
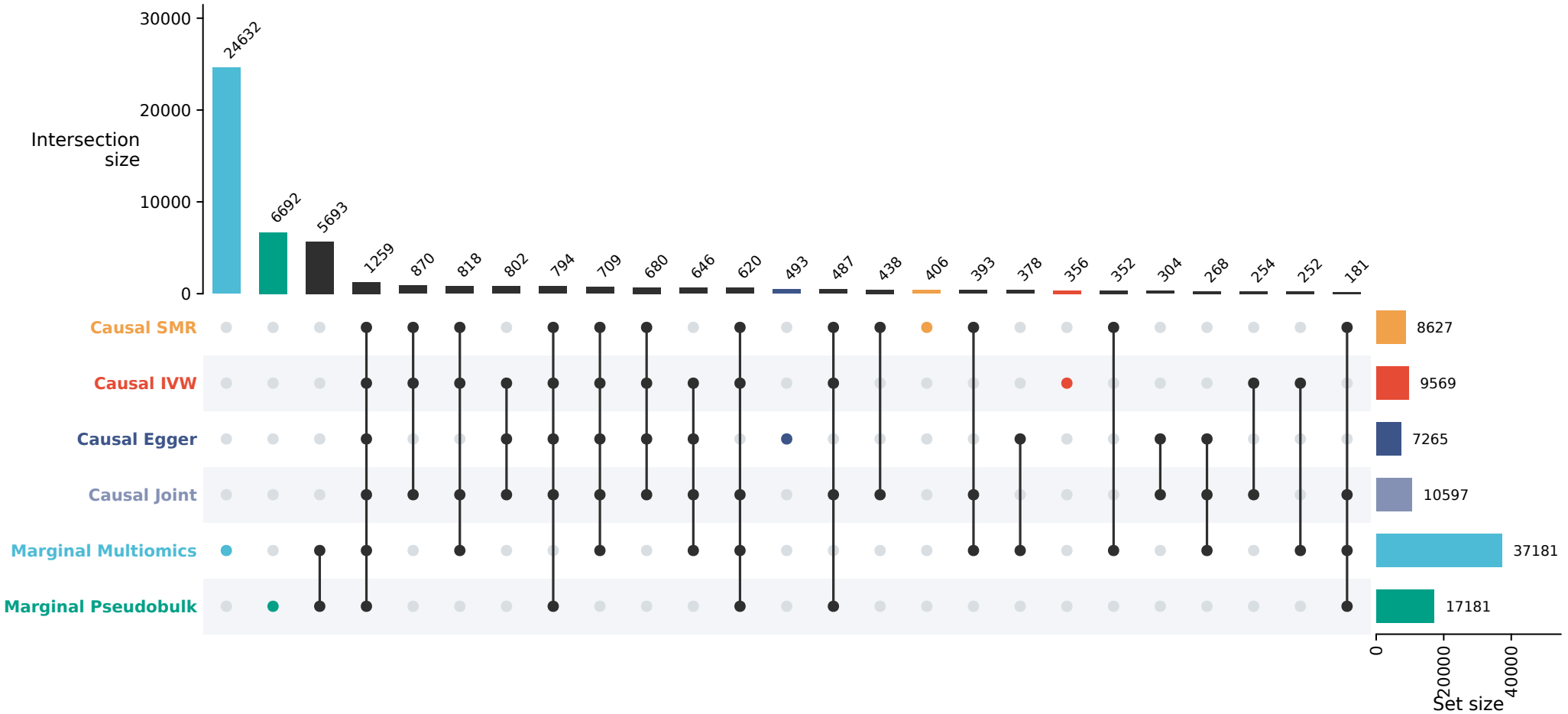
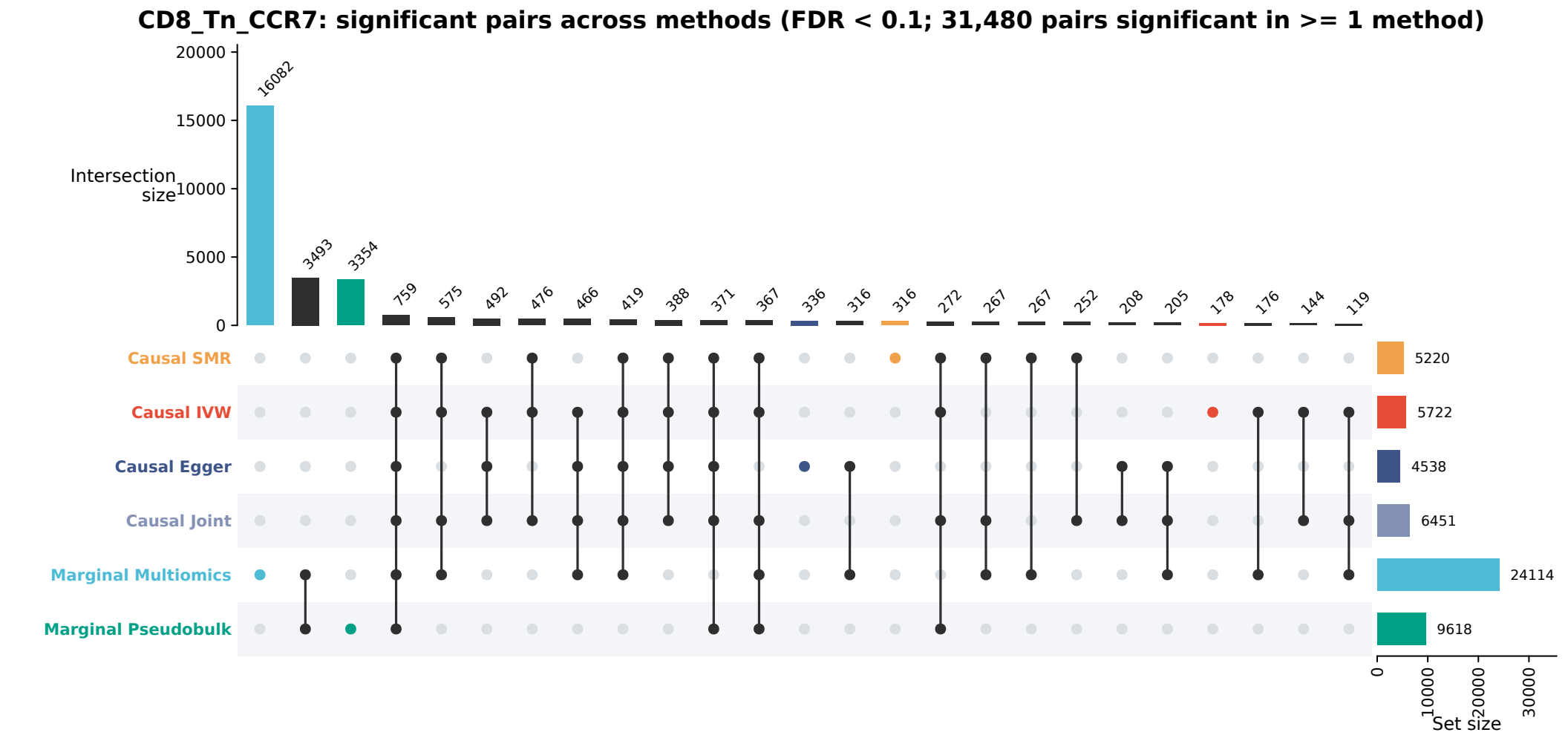
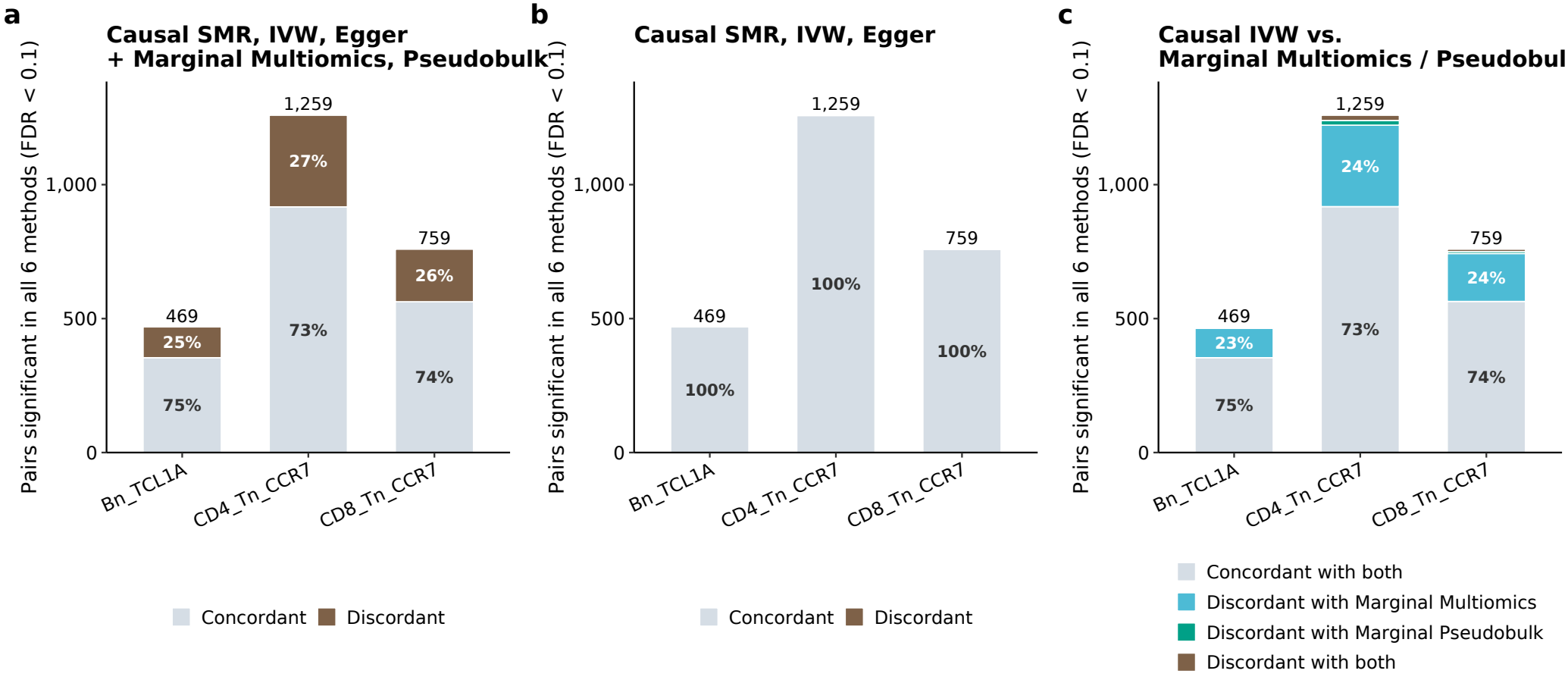

a
Bn_TCL1A (n = 40,087 pairs)
Significant pairs (FDR < 0.1)
0
5,000
10,000
15,000
20,000
3,534 (8.8%)
3,977 (9.9%)
3,361 (8.4%)
4,406 (11.0%)
18,179 (45.3%)
4,924 (12.3%)
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
b
CD4_Tn_CCR7 (n = 85,527 pairs)
Significant pairs (FDR < 0.1)
0
10,000
20,000
30,000
40,000
8,627 (10.1%)
9,569 (11.2%)
7,265 (8.5%)
10,597 (12.4%)
37,181 (43.5%)
17,181 (20.1%)
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
c
CD8_Tn_CCR7 (n = 50,931 pairs)
Significant pairs (FDR < 0.1)
0
10,000
20,000
5,220 (10.2%)
5,722 (11.2%)
4,538 (8.9%)
6,451 (12.7%)
24,114 (47.3%)
9,618 (18.9%)
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk


Bn_TCL1A: significant pairs across methods (FDR < 0.1; 22,766 pairs significant in >= 1 method)
Intersection size
15000
10000
5000
0
13427
1650
1587
469
432
360
340
338
311
306
280
270
265
244
236
217
179
175
161
158
157
153
132
105
98
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
3534
3977
3361
4406
18179
4924
0
10000
20000
Set size


CD4_Tn_CCR7: significant pairs across methods (FDR < 0.1; 50,704 pairs significant in >= 1 method)
Intersection size
30000
20000
10000
0
24632
6692
5693
1259
870
818
802
794
709
680
646
620
493
487
438
406
393
378
356
352
304
268
254
252
181
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
8627
9569
7265
10597
37181
17181
0
20000
40000
Set size


CD8_Tn_CCR7: significant pairs across methods (FDR < 0.1; 31,480 pairs significant in >= 1 method)
Intersection size
20000
15000
10000
5000
0
16082
3493
3354
759
575
492
476
466
419
388
371
367
336
316
316
272
267
267
252
208
205
178
176
144
119
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
5220
5722
4538
6451
24114
9618
0
10000
20000
30000
Set size

Effect direction among pairs significant in all six methods